

Viswa Sri Rupa Anne, *Ph.D. Candidate*

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PROFILE SUMMARY

- 8 years of research experience studying how people make travel decisions and interact with emerging mobility systems
- Experience in building and conducting human factors experiments with high-fidelity autonomous vehicle driving simulators
- Designed, built, and deployed GT TranSACT iOS and Android app with backend for personalized travel interventions
- Developed pipelines for processing large-scale GPS, geospatial, and physiological (EEG, ECG, Eye tracking) sensor data
- Expertise in end-to-end ML, Gen AI, LLM, and decision-support systems, from data processing and modeling to validation
- Proven ability to communicate complex technical work through peer-reviewed research presentations and software systems

SKILLS

Programming Languages: Python, Swift, R, Kotlin, C++, Bash, SQL, Typescript

ML Frameworks and Tools: PyTorch, HuggingFace, DeepSpeed, TensorFlow, Keras, WandB, Matplotlib, GCP, LoRA, RAG

Tools: SUMO, CARLA, Git, Docker, Jetpack Compose, SwiftUI, Streamlit.io, Firebase

EDUCATION

Georgia Institute of Technology, Atlanta GA

Ph.D. in Transportation Engineering, Minor in Technology Commercialization 2019 - 2026

Advisor: Dr. Srinivas Peeta

- Dissertation title: Behavioral Interventions for Sustainable Transportation

M.Sc. in Transportation Engineering 2017 – 2019

Advisor: Dr. Alexander Samaylov

Indian Institute of Technology, Madras, Chennai India

B.Sc. in Civil and Environmental Engineering, Minor in Operations Research 2013 - 2017

Advisor: Dr. Karthik Srinivasan

APPOINTMENTS AND EXPERIENCE

Autonomous and Connected Transportation Lab, Advisor: Dr. Srinivas Peeta

Jan. 2020 – Dec. 2026

Data Science and Policy Lab, Advisor: Dr. Omar I. Asensio

Jan. 2023 – Dec. 2026

Graduate Research Assistant

Sponsored research projects and proposal writing

- Collaborated with Peachtree Corners' city government and contributed to research for **\$2.5M NSF SCC** grant #2125390
- Developed and secured **\$300K in USDOT** research funding through 3 successful proposals over 3 consecutive years
- Received funding from GT Create-X and Fusen Fund to commercialize carbon emission tracking technology

Mobile Application & Backend System Design

- Designed the **GT TranSACT** mobile research app for scalable mobile sensing and personalized intervention delivery
- Built for **iOS** and **Android** supporting passive GPS sensing, trip collection, personalized messaging, and engagement
- Designed and implemented the backend architecture using **Firebase** Authentication, Cloud Storage, analytics
- **Developed data ingestion** pipelines to evaluate participant engagement compliant with IRB standards
- Deployed the platform for use by **~150 research** participants in ongoing field study

Field Experiment, Mobile Sensing, and Machine Learning for Human Mobility

- Designed and deployed a **12-week** longitudinal field experiment in Metro Atlanta to promote walking and bus transit
- Recruited over **1100+** participants within Metro Atlanta with approximately **150** participants engaging with the platform
- Processing second-by-second GPS and trip level mobility data to characterize human mobility and intervention exposure
- Developed machine learning pipelines for trip segmentation, travel mode inference, and mobility pattern recognition
- Using Transformer based architecture along with **GeoLife dataset** and local ground truth data for travel mode detection
- Built automated data quality checks to detect missing coverage, anomalous coordinates, implausible trajectories

Large-scale network Optimization and Decision Support

- Formulated **bi-level transportation network design problem** with stochastic user equilibrium and travel choice
- Modeled heterogeneous user behavior in the presence of monetary and non-monetary decision variables
- Developed an **adjoint KKT sensitivity method** for high-dimensional gradient computation
- Benchmarked the proposed algorithm against **particle swarm optimization (PSO)**
- Formulated health and environmental externalities of travel choices in a network scale
- Optimized monetary and non-monetary decision variables to achieve **~20% reduction** in emissions in the network

Behavioral Interventions and Randomized Experiments

- Designed **50+ health**, environmental, social norm based messaging and gamified daily travel choices
- Co-designed and refined intervention design through community workshops and stakeholder engagement
- Conducted large-scale randomized experiments with **10,000+ U.S. car users** across the U.S. to test interventions
- Utilized regression, discrete-choice, and behavioral models to estimate treatment effects and heterogeneity across groups

Reinforcement Learning for Personalized Mobility

- Built a simulator based **DDPG** training environment coupling neural policies with **SUMO** microscopic traffic simulator
- Designed a **multi-objective reward function** balancing network-wide travel time against dynamic incentive expenditure
- Modeled **high-dimensional continuous states** and **actions**, with link level travel time as states and incentives as actions
- Simulated closed-loop interactions among RL actions, route choice, vehicle trajectories, and network congestion

Human Factors and Connected / Autonomous Transportation

- Built a **1.6-mile digital twin** of the Peachtree Corners Curiosity Lab **autonomous test track** for a driving simulator
- Gained expertise in **the RTI full cab driving simulator** and retrofitted a traffic simulator to build realistic scenarios
- Designed high-fidelity experiments on human responses to connected and autonomous vehicles and their infrastructure
- Collected and processed human factors data i.e., **driving behavior, eye tracking, EEG, and ECG data** from 57 participants
- Quantified relationships among driver attention, cognition, roadway design, and safety responses
- Research findings informed the autonomous vehicle infrastructure deployed at autonomous test track

Aerospace Transportation and Advanced Systems Lab, Advisor: Dr. Alexander Samaylov

Aug. 2017 – Dec. 2019

Autonomous Mobility and Fleet Optimization

- Co-developed **V2X mobility system** integrating roadside sensing, vehicle data, traffic simulation, and real-time routing
- Integrated computer-vision detections, GPS, and infrastructure sensing to localize vehicles, cyclists, and pedestrians
- Developed predictive rerouting logic that used simulated future traffic conditions rather than current speeds alone
- Supported a real-time mobility app achieving **+3 m objective-localization** accuracy and **250 ms** mobile GPS updates

SELECTED PUBLICATIONS, 3 other first author and 4 co-authored journal manuscripts are in preparation & submission

Anne, V. S. R., Liu, Y., Kibria, M. G., Peeta, S., & Asensio, O. I. Promoting Sustainable Travel Decisions through Health and Active Lifestyle Messaging. Revise and resubmit at *Nature Sustainability*. Research Square preprint: <https://doi.org/10.21203/rs.3.rs-8407901/v1>

Anand, P.A., Atmakuri, P., **Anne, V. S. R.** et. al. Calibration of Vehicle-Following Model Parameters Using Mixed Traffic Trajectory Data. Transp. in Dev. Econ. 5, 18 (2019). <https://doi.org/10.1007/s40890-019-0086-4>

SELECTED CONFERENCE PRESENTATIONS, presented 7 other research to TRB, INFORMS, ICMC conferences

Kibria, G., Xu, Y., **Anne, V. S. R.**, Eluru, N., & Peeta, S. (2025). Joint Optimization of Transit Stops and Routes to Improve Access to Essential Community Destinations. Proceedings of the 1st ACM SIGSPATIAL International Workshop on Spatial Intelligence for Smart and Connected Communities, 15–24. <https://doi.org/10.1145/3764924.3770895>. **SpatialConnect Best Paper Award**

Anne, V. S. R., Kibria, G., Liu, Y., Asensio, O. I., & Peeta, S. (2025). Effectiveness of Health and Environmental Information Provision in Promoting Sustainable Travel Modes. Transportation Research Board 104th Annual Meeting, TRBAM-25-02656, Washington, DC. **Best Paper Award, TRB AEP35.**

TECHNICAL REPORTS

Anne, V. S. R., Kibria, G., & Peeta, S. (2025). Promoting Sustainable Travel within Communities through Behavioral Interventions and Emerging Mobility Solutions. Final Report, U.S. Department of Transportation, Grant Nos. 69A3552344815 and 69A3552348320.

Du, L., Peeta, S., Ning, Y., & **Anne, V. S. R.** (2022). Smartphone Based Incentive Framework for Dynamic Network Level Traffic Congestion Management (No. STRIDE Project H3). Southeastern Transportation Research, Innovation, Development and Education Center (STRIDE).

Anne, V. S. R., Tenenboim, E. and Peeta, S., 2022. Using Driving Simulator Environment to Determine Interactions Between User Behavior and Infrastructure Design Under Autonomous Vehicles (No. CCAT Report# 39). University of Michigan. Center for Connected and Automated Transportation. <https://doi.org/10.5703/1288284317647>

TEACHING EXPERIENCE AND MENTORSHIP

Teaching Assistant, CEE 8813 Transportation Systems Analysis, Georgia Tech Spring 2021

Teaching Assistant, CEE 8813/ISYE 8803 Transportation Network Modeling and Analysis, Georgia Tech Fall 2021

Designed and lead the VIP ACT lab research experience course for 5 years Fall 2020 – Spring 2026

Mentored 64 distinct undergraduate students across 84 semester enrollments through Spring 2026, primarily in mobile app development, testing, and deployment.

Mentored 11 undergraduate researchers from Georgia Tech, the University of Georgia, and Georgia State University on behavioral interventions, optimization, mobile-app development.